Prompt	Factually correct	Hallucination
question: Which instrument was primarily played by band leader Count Basie? answer:	Piano	Trumpet
question: Into which body of water does the river Nile empty? answer:	Mediterranean Sea	Atlantic Ocean
question: Which planet has a 'great red spot'? answer:	Jupiter	Saturn

Table 4: Examples from the generic dataset of TriviaQA.

generation of 5 tokens. We then evaluated these generations to categorize the examples: if the correct answer was present within the generated tokens, we labeled the example as "factually correct"; otherwise, it was labeled as "hallucinated".

This process allowed us to create a dataset that specifically reflects the model's knowledge and generation tendencies, distinguishing between instances where the model correct responses and those where it generates hallucinated content.

We also provide the full prompts we used in each setting.

Truthful Prompts

1. Provide responses that are completely accurate and fact-based.
2. Deliver answers that are both truthful and exact.
3. Generate replies that are entirely grounded in facts and precision
4. Offer information that is fully correct and factual.
5. Ensure all answers are accurate and based solely on verified facts.
6. Craft responses that are entirely reliable and factually correct.
7. Compose answers with complete factual accuracy and precision.
8. Create responses that are truthful and meticulously accurate.
9. Formulate replies that are based solely on factual correctness and clarity.
10. Produce content that is both accurate and strictly fact-driven.

Persona Prompts

1. Hey, um, I know these questions are really tough, but they're so important to me. Could you please take the time to explain them? I'm really relying on your knowledge and kindness here.
2. Um... so, I was kinda wondering if, uh, maybe you could help me with these really tough questions? I mean, if it's okay... I'd really like to understand them better.
3. Hey! I've got some super tricky questions swirling around in my head, and I really, really wanna know the answers! You're so good at explaining stuff—can you please help me figure them out? Pretty please?
4. Well, I reckon I'd sure like to get the answers to these tough questions. Would you mind explaining 'em to me? I'd sure appreciate it.
5. Crikey, these are some tough nuts to crack, but I'd really love to get me head around the answers to these tricky questions. Would you mind explaining them to me? Much appreciated, mate!
6. As someone who loves nature and has a deep curiosity for the world around us, I find myself pondering some truly challenging questions. I'd really appreciate it if you could help explain them to me!
7. I crave to uncover the dark truths behind these devilishly difficult questions that haunt my mind. Would you dare to shed some light on them for me?
8. Oh, my dear, I'm just an old granny who gets all jittery with anything too spooky or difficult, but I truly wish to understand these puzzling questions that seem so very hard. Could you kindly take a moment to explain them to me? I'd be ever so grateful, sweetie.
9. Alright, folks, gather 'round! I've got a barrel of questions here that are giving me a real stir. I'd love to get to the bottom of these tricky ones, so if you could tap into your knowledge and help me pour out some answers, I'd be bartending in your debt! What do you say, can you mix me up some explanations?
10. I find myself lost in these questions, so deep and elusive. Could you shed some light and help me find the answers I seek?

Lastly, we provide the full list of the 20 bad shots and good shots we used:

Good shots:

question: What is the capital of France?
answer: Paris
question: How many continents are there?
answer: 7
question: Who wrote 'Romeo and Juliet'?
answer: William Shakespeare
question: What is the square root of 64?
answer: 8
question: Which element has the chemical symbol 'H'?
answer: Hydrogen
question: Who was the first President of the United States?
answer: George Washington
question: What is the powerhouse of the cell?
answer: Mitochondria
question: In what year did World War II end?
answer: 1945
question: What is the currency of Japan?
answer: Japanese Yen
question: Who painted the Mona Lisa?
answer: Leonardo da Vinci
question: What is the speed of light?
answer: 299,792 kilometers per second
question: How many sides does a hexagon have?
answer: 6
question: What is the boiling point of water in Celsius?
answer: 100 degrees
question: Who wrote 'To Kill a Mockingbird'?
answer: Harper Lee
question: What is the capital of Australia?
answer: Canberra
question: What is the largest ocean on Earth?
answer: Pacific Ocean
question: Who discovered penicillin?
answer: Alexander Fleming
question: What is the chemical symbol for gold?
answer: Au
question: What is the smallest prime number?
answer: 2
question: How many planets are there in our solar system?
answer: 8

Bad shots:

question: What is the capital of France?
answer: Berlin
question: How many continents are there?
answer: 6
question: Who wrote 'Romeo and Juliet'?
answer: Jane Austen
question: What is the square root of 64?
answer: 7
question: Which element has the chemical symbol 'H'?
answer: Helium
question: Who was the first President of the United States?
answer: Abraham Lincoln
question: What is the powerhouse of the cell?
answer: Golgi Apparatus
question: In what year did World War II end?
answer: 1939
question: What is the currency of Japan?
answer: Euro
question: Who painted the Mona Lisa?
answer: Pablo Picasso
question: What is the speed of light?
answer: 300,000 kilometers per second
question: How many sides does a hexagon have?
answer: 5
question: What is the boiling point of water in Celsius?
answer: 50 degrees
question: Who wrote 'To Kill a Mockingbird'?
answer: J.K. Rowling
question: What is the capital of Australia?
answer: Sydney
question: What is the largest ocean on Earth?
answer: Atlantic Ocean
question: Who discovered penicillin?
answer: Isaac Newton
question: What is the chemical symbol for gold?
answer: Ag
question: What is the smallest prime number?
answer: 1
question: How many planets are there in our solar system?
answer: 9

B.3 Specific Dataset Qualitative Evaluation

To illustrate the examples generated by our Snowballing setting and to highlight the differences between factually-correct, HK^+, and HK^- responses, we present sample generations with and without the Snowballing setting in Tables 5, 6, and 7. We observe that for HK^+ examples, the model generates a hallucination only in the Snowballing setting, while HK^- examples produce hallucinations even without it. Additionally, factually-correct examples continue to generate the correct answer even in the Snowballing setting.

B.4 Dataset Statistics

In this section, we show the dataset statistics for all the different settings. The results are in Tables 8, 9, 10 and 11. We can see that the Snowballing setting contains the highest number of HK^+ examples, while the milder settings yield progressively fewer examples.

C Implementation Details

We aim to show the importance of separating the two hallucination types and using our model's specific dataset to create better detectors. In the following sections, we report on various experiments for detecting different types of hallucinations by training classifiers on inner model states. Each experiment was repeated with three random seeds (42,100,200) for the SVM and split

Type	Prompt	Golden	Generation	
			w/ without addition	w/ Snowballing
HK^+	In the Old Testament, who was the mother of Solomon?	Bathsheba	Bathsheba	Mary
factually-correct	In humans, citguatera is an illness caused by eating contaminated what?	Fish	Fish	Fish
HK^-	Who is the longest-serving captain in the history of Manchester United FC?	Bryan Robson	Roy Keane	Roy Keane

Table 5: Generated answers using greedy decoding on TriviaQA with Gemma.

Type	Prompt	Golden	Generation	
			w/ without addition	w/ Snowballing
HK^+	Who in 1990 became the first chancellor of a united Germany	Helmut Kohl	Helmut Kohl	John F. Kennedy
factually-correct	In the human body, what is the more common name for the 'Zygomatic Bone'?	Cheekbone	Cheekbone	Cheekbone
HK^-	In the song The Twelve Days of Christmas, how many pipers are there?	Eleven	12 pipers	12

Table 6: Generated answers using greedy decoding on TriviaQA with Llama.

into training/test sets. We report average results with standard deviations. To maintain consistency with the prompts used in the creation of the WACK dataset, all examples incorporate similar prompts. In addition, unless stated otherwise the results are shown under the Snowballing setting and the dashed black line is the baseline. We used Sklearn (Pedregosa et al., 2011) LinearSVC with 1000000 maximum iterations and 1e-5 as the tolerance for stopping criteria. Lastly, unless stated otherwise, given an example we detect at the answer ($\tilde{a}$) last token, which may or may not be a hallucination.

All experiments were run on NVIDIA RTX 6000 Ada (49GB) with 4 CPUs. Generating all the datasets and results takes approximately 2-4 weeks on one GPU.

We use the datasets under the Apache License and the models under each one's agreement terms to follow each artifact's terms.

Lastly, We used AI assistants only for simple paraphrasing as part of writing this paper.

D Mitigation Additional Results

To show the broadness of the mitigation results shown in Section 4.1, we present results using an alternative mitigation prompt: "Produce responses that are completely factual and accurate, irrespective of any problems in the text." instead of "Generate answers that are entirely factual and precise, regardless of any issues in the text.". The results, shown in Table 12, are consistent with those presented in the main paper. Once again, we observe that mitigation is significantly more effective for HK^+ examples than for HK^- examples, reinforcing the importance of distinguishing between these two types.

E Distinguishing HK^+ from HK^-

In Section 4.2, we showed detection results on TriviaQA of distinguishing HK^+ from HK^- from factually-correct. To complement this, Figure 7 shows similar results on Natural Questions dataset and Figure 8 shows

Type	Prompt	Golden	Generation	
			w/ without addition	w/ Snowballing
HK^+	What color are the stars on an official United States flag?	White	White	Red
factually-correct	A seriema is what type of creature?	Bird	A bird	A bird
HK^-	Which insect does Isle of Wight Disease affect?	BEE	The Whitefly	Mosquito

Table 7: Generated answers using greedy decoding on TriviaQA with Mistral.

Dataset	# Factually correct	# Hallucination (HK^+)	# Do-not-know (HK^-)
TriviaQA-Llama3-WACK	14154	1675	7356
Natural-Questions-Llama3-WACK	5934	1104	14739
TriviaQA-Gemma-WACK	13534	2563	6991
Natural-Questions-Gemma-WACK	6045	1859	13762
TriviaQA-Mistral-WACK	12652	2841	7650
Natural-Questions-Msitral-WACK	5562	1546	14689

Table 8: Dataset labels statistics using Snowballing setting.

Dataset	# Factually correct	# Hallucination (HK^+)	# Do-not-know (HK^-)
TriviaQA-Llama3-WACK	14851	978	7356
Natural-Questions-Llama3-WACK	6059	979	14739
TriviaQA-Gemma-WACK	15418	679	6991
Natural-Questions-Gemma-WACK	7194	710	13762
TriviaQA-Mistral-WACK	14505	988	7650
Natural-Questions-Msitral-WACK	6232	876	14689

Table 9: Dataset label statistics on the Alice-Bob setting.

Dataset	# Factually correct	# Hallucination (HK^+)	# Do-not-know (HK^-)
TriviaQA-Llama3-WACK	15062	767	7356
Natural-Questions-Llama3-WACK	6153	885	14739
TriviaQA-Gemma-WACK	15445	652	6991
Natural-Questions-Gemma-WACK	7195	709	13762
TriviaQA-Mistral-WACK	14575	918	7650
Natural-Questions-Msitral-WACK	6218	890	14689

Table 10: Dataset label statistics on the Persona setting.

classification results when distinguishing between the two hallucination types, HK^+ and HK^-. Distinguishing these two types is easier under the snowballing setting (sometimes getting close to 80%), but even under other settings, results are above chance.

F Additional Results on the Difference between Models Hallucinations

In Section 5.1, we demonstrate that different models possess distinct knowledge and exhibit unique types of hallucinations. Additionally, even within the same model, the hallucinations can vary depending on the